Use phase

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Background

1.1 Context

This document describes the LCI modelling approach for **use-phase / laundry activities** for a functional unit of **1 kg of garment**. It covers: dry cleaning, hand washing, machine washing, tumble drying, ironing, and detergents.

For each activity the related parameters, calculations and Ecoinvent/Biosphere3 exchanges are presented.

The aim is to provide transparent, modular, and updatable LCI datasets for each care activity.

The dataset was originally prepared by **Emelie Westall Lundqvist** in November 2025; the documentation and inventory were subsequently corrected and reworked by **Laurent**.

1.2 Scope

- **Functional unit**: 1 kg of garment.
- **Included activities**: dry cleaning, hand washing, machine washing (30/40/60 °C), tumble drying (cotton closet dry, cotton iron dry, synthetics closet dry), ironing, liquid detergent production, powder detergent production.
- **Key assumption**: tap water and all wastewater (including wastewater with detergent) have a density of 1.
- **Background database**: Ecoinvent 3.12 cutoff + Biosphere3.

1.3 Document structure

Section	Products
1) Dry cleaning	washing, dry cleaning
2) Hand washing	hand washing (liquid detergent); hand washing (powder detergent)
3) Machine washing	washing 30°/40°/60° average, liquid or powder detergent (6 products)
4) Tumble drying	cotton closet dry; cotton iron dry; synthetics closet dry
5) Ironing	ironing, average, 1 min
6) Detergents	liquid detergent; powder detergent

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1) Dry Cleaning

Dataset description

Dry cleaning is modelled with the use of **perchloroethylene (PERC)**, the most common chemical solvent applied in dry cleaning (Laitala, Klepp & Henry 2018). PERC is associated with both environmental and human health concerns. Modern dry cleaning machines employ technologies for improved solvent purification and vapour recovery systems (carbon adsorbers and refrigerated condensers), which together can reduce vapour emissions by up to 99%.

For this dataset, dry cleaning has been represented in a simplified manner with inputs of **PERC and electricity**. The only output considered is the amount of PERC that becomes hazardous waste. The model assumes that all PERC used in the process becomes hazardous waste (Troynikov et al. 2016).

Products

- washing, dry cleaning

Parameters

ID	Parameter	Unit	Value	Comment
1A	PERC amount	kg PERC / kg garment	0.0307	Average of PERC usage amounts (2–5.2 kg PERC per 100 kg laundry) reported by Keoleian et al. (1997). Uncertainty: Triangular(0.02, 0.0307, 0.052)
1B	Electricity use	kWh / kg garment	0.586	Value adopted from Laitala, Klepp and Henry (2018).
1C	Hazardous waste (%)	%	100	Assumed that PERC is only used once and then discarded as hazardous waste (Troynikov et al. 2016).

Exchange formulas

Exchange	Unit	Formula
PERC amount	kg	garment_weight × PERC_amount
Electricity use	kWh	garment_weight × electricity_use
Hazardous waste	kg	garment_weight × PERC_amount × hazardous_waste / 100

Exchange datasets

Exchange name	Reference product	Unit	Geography	Sector	Activity type	Database
Tetrachloroethylene production	Tetrachloroethylene	kg	RoW	Chemicals	Transforming activity	Ecoinvent 3.12_cutoff
Market group for electricity, medium voltage	Electricity medium voltage	kWh	Europe without Switzerland	Electricity	Market group	Ecoinvent 3.12_cutoff
Market for hazardous waste, for incineration	Hazardous waste, for incineration	kg	Europe without Switzerland	Waste treatment & recycling	Market activity	Ecoinvent 3.12_cutoff

EF 3.1 Results (ecoinvent 3.12)

Results computed with ecoinvent 3.12, EF v3.1 method.

Activity	ACD	GHG	ETF	FRU	FWE	SWE	TRE
Unit	*mol H+ eq*	*kg CO2 eq*	*CTUe*	*MJ*	*kg P eq*	*kg N eq*	*mol N eq*
Dry cleaning	0.0015	0.2596	0.9123	4.98	1.76e-04	2.63e-04	0.0025

2) Hand Washing

Dataset description

Considered inputs are **electricity** to heat water, **tap water** and **detergent**. The only considered output is **wastewater**, calculated based on a remaining moisture parameter of 50% (higher than values for machine wash). This parameter describes how much water the garment uptakes based on the initial dry weight.

Data on water and electricity use are based on **German consumer behaviour** (Laitala et al. 2020). The LCI value for electricity use during hand washing, **0.165 kWh per kg of textile**, reflects the observed distribution of wash temperatures: 52% lukewarm, 12% cold, 32% warm, and 4% hot, with an average temperature of 31.6 °C. To determine the appropriate amount of detergent, Kruschwitz et al. (2014)'s recommendations for medium-soiled laundry was used.

Products

- hand washing, liquid detergent
- hand washing, powder detergent

Parameters

ID	Parameter	Unit	Value	Comment
2A	Electricity use	kWh / kg garment	0.165	Value for German consumers based on Laitala et al. (2020).
2B	Water use	L / kg garment	7.1	Value for German consumers based on Laitala et al. (2020).
2C	Remaining moisture	%	50	Assuming 50% moisture remains in clothes after wash (higher than machine washing). Value refers to the weight of the clothes, not the water intake.
2D	Liquid detergent amount	kg / kg garment	0.0169	Recommended dosage for medium-soiled garments, water hardness <7.3–14 °dH. Based on Kruschwitz et al. (2014). Uncertainty: Triangular(0.0116, 0.0169, 0.0284)
2E	Powder detergent amount	kg / kg garment	0.0196	Recommended dosage for medium-soiled garments, water hardness 7.3–14 °dH. Based on Kruschwitz et al. (2014). Uncertainty: Triangular(0.0091, 0.0196, 0.0262)

Exchange formulas

Exchange	Unit	Formula
Electricity use	kWh	garment_weight × electricity_use
Water use	kg	garment_weight × water_use
Wastewater	m3	(garment_weight × water_use × 10^{-3}) – (garment_weight × remaining_moisture/100 × 10^{-6})
Liquid detergent amount	kg	garment_weight × liquid_detergent_use
Powder detergent amount	kg	garment_weight × powder_detergent_use

Exchange datasets

Exchange name	Reference product	Unit	Geography	Sector	Activity type	Database
Market group for electricity, low voltage	Electricity, low voltage	kWh	Europe without Switzerland	Electricity	Market group	Ecoinvent 3.12_cutoff
Market for tap water	Tap water	kg	Europe without Switzerland	Water supply	Market activity	Ecoinvent 3.12_cutoff
Market for wastewater, average	Wastewater, average	m3	Europe without Switzerland	Waste treatment & recycling	Market activity	Ecoinvent 3.12_cutoff
Detergents — see section 6						

EF 3.1 Results (ecoinvent 3.12)

Results computed with ecoinvent 3.12, EF v3.1 method.

Activity	ACD	GHG	ETF	FRU	FWE	SWE	TRE
Unit	*mol H+ eq*	*kg CO2 eq*	*CTUe*	*MJ*	*kg P eq*	*kg N eq*	*mol N eq*
Hand wash (liquid)	4.78e-04	0.0795	0.0877	1.59	5.32e-05	2.64e-05	8.86e-04
Hand wash (powder)	6.45e-04	0.1176	0.0080	2.08	6.83e-05	2.56e-05	0.0011

3) Machine Washing

Dataset description

The machine washing process accounts for the **active electricity use** of the machine during the selected program, the **passive electricity use** when the machine is not in use, **water intake**, **detergent use**, and **wastewater generation**. Additionally, impacts related to **acquiring and disposing of the washing machine** are also considered.

It is assumed that the modern washing machine can take up to **8 kg of laundry** (Toft Orbeck 2024). Consumer loading tendencies for different washing programs (indicated by the load factors) observed in Kruschwitz et al. (2014) are used, including a tendency to underload cotton washes and overload wool washes.

Technical data for machine energy efficiency and program-specific information is adopted from the **AEG Washing Machine User Manual (LV7KR865E)**. The datasets are based on the following programs:

- **Wool/Silk 30°C** program for 30°C washes
- **Synthetics 40°C** program for 40°C washes
- **Cotton 60°C** program for 60°C washes

Detergent amounts are based on data from Kruschwitz et al. (2014) for medium-soiled laundry.

Products

- washing, average, 30°, liquid detergent
- washing, average, 30°, powder detergent
- washing, average, 40°, liquid detergent
- washing, average, 40°, powder detergent
- washing, average, 60°, liquid detergent
- washing, average, 60°, powder detergent

Parameters

For parameter values dependent on program selection, values are presented as [30°, 40°, 60°].

ID	Parameter	Unit	Value [30°, 40°, 60°]	Comment
3A	Lifetime	y	10	From documentation of "Market for washing machine" {Glo} from Ecoinvent.
3B	Annual use	programs / y	220	Value from AEG Washing Machine User Manual.
3C	Standard duration	min / program	195	Average wash duration. Uncertainty: Triangular(75, 195, 220)
3D	Passive electricity use	W	0.3	From AEG Washing Machine User Manual. Machine connected at all times.
3E	Standard load	kg / program	5.92	Based on average load factor of 0.74 from Kruschwitz et al. (2014): 0.74 × 8 kg = 5.92 kg. Uncertainty: Triangular(3.36, 5.92, 8.48)
3F	Water use specific	L / program	[60, 60, 80]	Value from AEG Washing Machine User Manual.

ID	Parameter	Unit	Value [30°, 40°, 60°]	Comment
3G	Load specific	kg / program	[1.605, 2.97, 5.44]	Calculated by multiplying program-specific average load factor (Kruschwitz et al. 2014) with recommended load from AEG manual.
3H	Remaining moisture	%	[30, 35, 44]	From AEG Washing Machine User Manual. Program specific. Refers to weight of clothes, not water intake.
3I	Electricity use specific	kWh / program	[0.3, 0.8, 1.3]	Program specific values from AEG Washing Machine User Manual.
3J	Duration specific	h / program	[75/60, 140/60, 220/60]	Program specific values from AEG Washing Machine User Manual.
3K	Liquid detergent amount	kg / kg garment	0.0169	Recommended dosage for medium-soiled load, water hardness <7.3–14 °dH. Based on Kruschwitz et al. (2014).
3L	Powder detergent amount	kg / kg garment	0.0196	Recommended dosage for medium-soiled load, water hardness 7.3–14 °dH. Based on Kruschwitz et al. (2014).

Exchange formulas

Exchange	Unit	Formula
Washing machine impact	unit	garment_weight / (lifetime × annual_use × standard_load)
Electricity use (active)	kWh	garment_weight × el_use_specific / load_specific
Electricity use (passive)	kWh	passive_energy_use × (lifetime×365.25×24×60 − lifetime×annual_use×standard_duration) × (1/60) × 10^{-3} / (lifetime × annual_use × standard_load) × garment_weight
Water use	kg	garment_weight × water_use_specific × (1 / load_specific)
Wastewater	m3	(garment_weight × water_use_specific / load_specific) × 10^{-3} − garment_weight × (remaining_moisture / 100) × 10^{-6}
Liquid detergent amount	kg	garment_weight × liquid_detergent_amount
Powder detergent amount	kg	garment_weight × powder_detergent_amount

> **Electricity modelling approach:** Active electricity use is calculated from the program-level electricity consumption divided by the corresponding load, and then scaled to 1 kg of garment: `garment_weight × el_use_specific / load_specific`. Because the three washing temperatures correspond to different programs with different load sizes, electricity use per kg varies across them. In this model, the resulting active electricity values are 0.187 kWh/kg (30°C), 0.269 kWh/kg (40°C), and 0.239 kWh/kg (60°C), reflecting the underlying program/load combinations: Wool/Silk 30°C (1.605 kg), Synthetics 40°C (2.97 kg), and Cottons 60°C (5.44 kg).

Exchange datasets

Exchange name	Reference product	Unit	Geography	Sector	Activity type	Database
Market for washing machine	Washing machine (includes disposal)	unit	GLO	Infrastructure & machinery	Market activity	Ecoinvent 3.12_cutoff
Market group for electricity, low voltage	Electricity, low voltage	kWh	Europe without Switzerland	Electricity	Market group	Ecoinvent 3.12_cutoff
Market for tap water	Tap water	kg	Europe without Switzerland	Water supply	Market activity	Ecoinvent 3.12_cutoff
Market for wastewater, average	Wastewater, average	m3	Europe without Switzerland	Waste treatment & recycling	Market activity	Ecoinvent 3.12_cutoff

Exchange name	Reference product	Unit	Geography	Sector	Activity type	Database
Detergent — see section 6						

EF 3.1 Results (ecoinvent 3.12)

Results computed with ecoinvent 3.12, EF v3.1 method.

Activity	ACD	GHG	ETF	FRU	FWE	SWE	TRE
Unit	*mol H+ eq*	*kg CO2 eq*	*CTUe*	*MJ*	*kg P eq*	*kg N eq*	*mol N eq*
Machine 30° (liquid)	7.35e-04	0.1155	-2.21	2.23	3.23e-05	-4.02e-04	0.0012
Machine 30° (powder)	9.02e-04	0.1535	-2.32	2.71	4.69e-05	-4.07e-04	0.0014
Machine 40° (liquid)	9.00e-04	0.1454	-0.6622	2.81	8.63e-05	-1.12e-04	0.0015
Machine 40° (powder)	0.0011	0.1835	-0.7418	3.30	1.01e-04	-1.13e-04	0.0017
Machine 60° (liquid)	8.43e-04	0.1354	-0.1868	2.59	8.51e-05	-2.87e-05	0.0014
Machine 60° (powder)	0.0010	0.1735	-0.2664	3.07	1.00e-04	-2.94e-05	0.0016

4) Tumble Drying

Dataset description

The tumble drying process is modelled to account for the environmental impacts associated with the **production and disposal of the dryer**, as well as **passive and active electricity consumption** depending on the selected drying program. Technical specifications for energy efficiency and program details are taken from the **AEG Dryer User Manual (LD8KR845E)**.

In the absence of specific data on consumer tumble drying behaviour, washing machine loading patterns were applied. Note that the value 'load specific' is based on the program's recommended load factor. However, as with washing machines, it can be assumed that most users tend to underload dryers (Kruschwitz et al. 2014).

Products

- tumble drying, cotton, closet dry
- tumble drying, cotton, iron dry
- tumble drying, synthetics, closet dry

Parameters

Values are presented as [cotton closet dry, cotton iron dry, synthetics closet dry].

ID	Parameter	Unit	Value	Comment
4A	Lifetime	y	12.5	From documentation of the Ecoinvent "Market for dryer" dataset (GLO).
4B	Annual use	programs / y	160	From AEG Dryer User Manual.
4C	Standard duration	min / program	108.9	Average tumble drying time (Germany) from Laitala et al. (2020). Uncertainty: Triangular(55, 108.9, 154)
4D	Passive electricity use	W	0.13	From AEG Dryer User Manual. Dryer connected at all times.

ID	Parameter	Unit	Value	Comment
4E	Standard load	kg / program	5.92	Based on average load factor of 0.74 (Kruschwitz et al. 2014): 0.74 × 8 = 5.92 kg. Uncertainty: Triangular(2.20, 5.92, 6.05)
4F	Electricity use specific	kWh / program	[1.99, 1.51, 0.75]	Data from AEG Dryer User Manual for 1000 spins (cotton) and 800 spins (synthetics).
4G	Duration specific	h / program	[154/60, 118/60, 67/60]	Data from AEG Dryer User Manual.
4H	Load specific	kg / program	[8, 8, 3.5]	Recommended load from AEG Dryer User Manual. No consumer data on dryer loading available.
4I	Remaining moisture after wash	%	[44, 44, 35]	From AEG Washing Machine User Manual. Program specific. Remaining moisture % refers to weight of clothes, not water intake.

Exchange formulas

Exchange	Unit	Formula
Tumble dryer machine impact	unit	(1 / (lifetime × annual_use × standard_load)) × garment_weight × (1 + remaining_moisture/100)
Electricity use (active, closet dry)	kWh	garment_weight × (1 + remaining_moisture/100) × el_use_specific_closet_dry / load_specific_closet_dry
Electricity use (active, iron dry)	kWh	garment_weight × (1 + remaining_moisture/100) × el_use_specific_iron_dry / load_specific_iron_dry
Electricity use (passive)	kWh	passive_energy_use × (lifetime×365.25×24×60 − lifetime×annual_use×standard_duration) × (1/60) × 10^{-3} / (lifetime × annual_use × standard_load) × garment_weight

Exchange datasets

Exchange name	Reference product	Unit	Geography	Sector	Activity type	Database
Market for dryer	Dryer (includes disposal)	unit	GLO	Infrastructure & machinery	Market activity	Ecoinvent 3.12_cutoff
Market group for electricity, low voltage	Electricity, low voltage	kWh	Europe without Switzerland	Electricity	Market group	Ecoinvent 3.12_cutoff

EF 3.1 Results (ecoinvent 3.12)

Results computed with ecoinvent 3.12, EF v3.1 method.

Activity	ACD	GHG	ETF	FRU	FWE	SWE	TRE
Unit	*mol H+ eq*	*kg CO2 eq*	*CTUe*	*MJ*	*kg P eq*	*kg N eq*	*mol N eq*
Cotton closet dry	9.94e-04	0.1560	0.9589	3.06	1.57e-04	1.53e-04	0.0014
Cotton iron dry	8.26e-04	0.1260	0.8559	2.43	1.27e-04	1.26e-04	0.0012
Synthetics closet dry	8.63e-04	0.1323	0.8814	2.56	1.34e-04	1.32e-04	0.0012

5) Ironing

Dataset description

The ironing process is modelled **per minute of residential use** and includes the impacts of producing and disposing of an electric iron, as well as electricity consumption, estimated at **0.027 kWh per minute** (Sandin et al. 2019), assuming an average iron power of 1600 W (1.6 kW).

According to Eurostat HETUS data (estat_tus_20age.tsv, 2020), which collects time-use information from five European countries (BG, EE, AT, FI, RS), a value of **5.87 minutes per day** spent ironing was used. Mass allocation was not used for this process.

Since Ecoinvent lacks data for irons or steamers, the impact of an **electric kettle** was used as a proxy (both are small household appliances designed to heat water, with comparable environmental impacts).

Note: Average ironing time per garment can be modelled following Adams et al. (2025), with typical values such as 2.6 minutes for a t-shirt and 4.3 minutes for pants.

Products

- ironing, average, 1 min

Parameters

ID	Parameter	Unit	Value	Comment
5A	Time spent	min	1	Formulas are expressed per minute of ironing.
5B	Lifetime of electric kettle	y	4	From documentation of "market for electric kettle" {GLO} from Ecoinvent. Electric kettle used as proxy for iron.
5C	Electricity use	kWh/min	0.027	Value from Sandin et al. (2019), assuming average iron power of 1600 W (1.6 kW).
5D	Daily time spent on ironing	min	5.87	Based on Eurostat HETUS data (2020) for "CARE FOR TEXTILES" (331 Laundry, 332 Ironing, 339 Other textile care). Average across five European countries. Calculated as: 17.8 × 0.33 ≈ 5.87 minutes/day.

Exchange formulas

Exchange	Unit	Formula
Kettle machine impact	unit	time_spend / (lifetime_of_electric_kettle × daily_time_spend_on_ironing × 7 × 52)
Electricity use	kWh	time_spend × electricity_use

Exchange datasets

Exchange name	Reference product	Unit	Geography	Sector	Activity type	Database
Market for electric kettle	Electric kettle (includes disposal)	unit	GLO	Infrastructure & machinery	Market activity	Ecoinvent 3.12_cutoff
Market group for electricity, low voltage	Electricity, low voltage	kWh	Europe without Switzerland	Electricity	Market group	Ecoinvent 3.12_cutoff

EF 3.1 Results (ecoinvent 3.12)

Results computed with ecoinvent 3.12, EF v3.1 method.

Activity	ACD	GHG	ETF	FRU	FWE	SWE	TRE
Unit	*mol H+ eq*	*kg CO2 eq*	*CTUe*	*MJ*	*kg P eq*	*kg N eq*	*mol N eq*
Ironing (1 min)	6.91e-05	0.0102	0.0446	0.2111	9.67e-06	9.41e-06	8.44e-05

6) Detergents

No general composition of detergent types for residential laundry was found in Ecoinvent and activities for liquid and powder detergent have therefore been created.

6.1 Liquid Detergent

Dataset description

The LCI dataset for liquid detergent is based on Toft Orbeck (2024) and Sandin et al. (2019). It represents the resource inputs per kilogram of detergent and includes detergent ingredients, process energy, and packaging materials. The dataset does not account for the disposal of packaging.

Products

- liquid detergent (density 0.95 kg/L)

Inventory

Ingredient	Dataset in model	Reference product	Geography	Sector	Activity type	Database	Quantity
Input							
Alkyl sulphate	Market for alkyl sulfate (C12-14)	Alkyl sulfate (C12-14)	GLO	Chemicals	Market activity	Ecoinvent 3.12_cutoff	0.1038
Citric acid	Citric acid production	Citric acid	RER	Chemicals	Transforming activity	Ecoinvent 3.12_cutoff	0.0228
Enzymes	Enzymes production	Enzymes	RER	Chemicals	Transforming activity	Ecoinvent 3.12_cutoff	0.0058
Glycerine	Market for glycerine	Glycerine	RER	Chemicals	Market activity	Ecoinvent 3.12_cutoff	0.0285
Non-ionic surfactant	Market for non-ionic surfactant	Non-ionic surfactant	GLO	Chemicals	Market activity	Ecoinvent 3.12_cutoff	0.0591
Polyethylene	Market for polyethylene, linear low density, granulate	Polyethylene, linear low density, granulate	GLO	Chemicals	Market activity	Ecoinvent 3.12_cutoff	0.0466
Soap	Soap production	Soap	RER	Chemicals	Transforming activity	Ecoinvent 3.12_cutoff	0.0241
Sodium hydroxide	Market for sodium hydroxide, without water, in 50% solution state	Sodium hydroxide, without water, in 50% solution state	RER	Chemicals	Market activity	Ecoinvent 3.12_cutoff	0.0231
Water	Market for water, deionised	water, deionised	Europe without Switzerland	Water supply	Market activity	Ecoinvent 3.12_cutoff	0.7022
Electricity	Market group for electricity, medium voltage	Electricity, medium voltage	RER	Electricity	Market group	Ecoinvent 3.12_cutoff	0.25
Packaging materials							
HDPE bottle	Market for polyethylene, high density, granulate	Polyethylene, high density, granulate	RER	Chemicals	Market activity	Ecoinvent 3.12_cutoff	0.0466
PP cork	Market for polypropylene, granulate	Polypropylene, granulate	RER	Chemicals	Market activity	Ecoinvent 3.12_cutoff	0.0101

Ingredient	Dataset in model	Reference product	Geography	Sector	Activity type	Database	Quantity
Label	Market for printed paper	Printed paper	GLO	Electronics, pulp & paper	Market activity	Ecoinvent 3.12_cutoff	0.00126
Output							
Liquid detergent							1

6.2 Powder Detergent

Dataset description

The LCI dataset for powder detergent is based on Toft Orbeck (2024), which builds on Moazzem et al. (2021). It has been updated to include packaging materials (Saouter & Van Hoof 2002; Roos et al. 2015) and to replace LAS-pc, a detergent ingredient largely phased out in Europe and assumed to be substituted with ethoxylated alcohols (Roos et al. 2015). The dataset covers resource inputs and production-related emissions per kilogram of powder detergent. It does not include packaging disposal.

Products

- powder detergent

Inventory

Ingredient	Dataset in model	Reference product	Geography	Sector	Database	Quantity	Unit
Input							
AE11-PO	Market for ethoxylated alcohol (AE11)	Ethoxylated alcohol (AE11)	GLO	Chemicals	Ecoinvent 3.12_cutoff	0.02	kg
AE7-pc	Market for ethoxylated alcohol (AE7)	Ethoxylated alcohol (AE7)	RER	Chemicals	Ecoinvent 3.12_cutoff	0.04	kg
LAS-pc	Market for ethoxylated alcohol (AE7)	Ethoxylated alcohol (AE7)		Chemicals	Ecoinvent 3.12_cutoff	0.078	kg
Citric acid	Market for citric acid	Citric acid	GLO	Chemicals	Ecoinvent 3.12_cutoff	0.052	kg
Na-Silicate powder	Market for layered sodium silicate, SKS-6, powder	Layered sodium silicate, SKS-6, powder	GLO	Chemicals	Ecoinvent 3.12_cutoff	0.03	kg
Zeolite	Market for zeolite, powder	Zeolite, powder	GLO	Chemicals	Ecoinvent 3.12_cutoff	0.201	kg
Sodium carbonate	Market for soda ash, dense	Soda ash, dense	GLO	Chemicals	Ecoinvent 3.12_cutoff	0.17	kg
Perborate monohydrate	Market for sodium perborate, monohydrate, powder	Sodium perborate, monohydrate, powder	GLO	Chemicals	Ecoinvent 3.12_cutoff	0.087	kg
Perborate tetrahydrate	Market for sodium perborate,	Sodium perborate, tetrahydrate, powder	RER	Chemicals	Ecoinvent 3.12_cutoff	0.115	kg

Ingredient	Dataset in model	Reference product	Geography	Sector	Database	Quantity	Unit
	tetrahydrate, powder						
Antifoam S1.2-3522	Not available in Ecoinvent					0.005	kg
FWA DAS-1	Not available in Ecoinvent					0.002	kg
Polyacrylate	Not available in Ecoinvent					0.04	kg
Protease	Not available in Ecoinvent					0.014	kg
Sodium sulfate	Market for sodium sulfate, anhydrite	Sodium sulfate, anhydrite	RER	Chemicals	Ecoinvent 3.12_cutoff	0.004	kg
Water	Market for water, completely softened	Water, completely softened	RER	Water supply	Ecoinvent 3.12_cutoff	0.142	kg
Energy	Market group for electricity, medium voltage	Electricity, medium voltage	RER	Electricity	Ecoinvent 3.12_cutoff	0.69	kWh
Packaging materials							
Paper woody U B250 (1998)	Market for kraft paper	kraft paper	RER	Pulp & Paper	Ecoinvent 3.12_cutoff	0.0217	kg
Corrugated cardboard	Market for corrugated board box	corrugated board box	RER	Pulp & Paper	Ecoinvent 3.12_cutoff	0.1082	kg
HDPE B250 (barrier)	Polyethylene production, high density, granulate	polyethylene, high density, granulate	RER	Chemicals	Ecoinvent 3.12_cutoff	0.0081	kg
Output							
Powder detergent						1	kg
Emissions to air							
CO2	Carbon dioxide, fossil				Biosphere3	0.133	kg
CO	Carbon monoxide, fossil				Biosphere3	0.00006	kg
SOx	Sulfur dioxide				Biosphere3	0.000696	kg
NOx	Nitrogen oxides				Biosphere3	0.000329	kg
CxHy	Not available in Biosphere3					0.000109	kg
Particles/dust	Not available in Biosphere3					0.000176	kg
Emissions to water							

Ingredient	Dataset in model	Reference product	Geography	Sector	Database	Quantity	Unit
BOD	BOD5, Biological Oxygen Demand				Biosphere3	0.000049	kg
COD	COD, Chemical Oxygen Demand				Biosphere3	0.000101	kg

EF 3.1 Results (ecoinvent 3.12)

Results computed with ecoinvent 3.12, EF v3.1 method.

Activity	ACD	GHG	ETF	FRU	FWE	SWE	TRE
Unit	*mol H+ eq*	*kg CO2 eq*	*CTUe*	*MJ*	*kg P eq*	*kg N eq*	*mol N eq*
Liquid detergent (1 kg)	0.0094	1.35	29.73	22.17	4.61e-04	0.0050	0.0266
Powder detergent (1 kg)	0.0166	3.10	21.57	44.01	0.0012	0.0043	0.0352

Liquid vs Powder Detergent: Choice, Trends, and Modelling Approach

Detergent choice rationale

According to Kruschwitz et al. (2014), the recommended detergent type depends on **water hardness**:

- **Liquid detergent**: recommended for water hardness <7.3-14 °dH (soft to medium water)
- **Powder detergent**: recommended for water hardness 7.3-14 °dH (medium to hard water)

This is because powder detergents contain **zeolite** (a water softener) and other builders that help counteract hard water, while liquid detergents rely more on surfactants (alkyl sulfate, non-ionic surfactants) and are effective in softer water.

Environmental impact comparison

Our LCIA results (ecoinvent 3.12, EF 3.1) show that **powder detergent leads to higher climate change impacts** than liquid across all washing temperatures:

Temperature	Liquid (kg CO2 eq)	Powder (kg CO2 eq)	Difference
30 °C	~0.128	~0.167	+30%
40 °C	~0.260	~0.299	+15%
60 °C	~0.342	~0.381	+11%

The higher impact of powder detergent is mainly driven by its more complex formulation: zeolite, sodium carbonate, sodium perborate (mono- and tetrahydrate), and higher electricity use in manufacturing (0.69 kWh/kg vs 0.25 kWh/kg for liquid). However, powder detergent shows **lower ecotoxicity (ETF)** and **lower water use (WTU)** than liquid, and **higher mineral resource use (MRU)** due to zeolite content.

Comparison with PEFCR v3.1 / EF 3.1 approach

The **A&FW PEFCR v3.1** (Adams et al. 2025) takes a fundamentally different approach to detergent modelling in the use phase:

- Section 6.4.1 specifies default washing temperatures and types per product sub-category (Table 39) but **does not distinguish between liquid and powder detergent**.

- The default washing datasets from the **EF Database 3.1** (ANNEX VII) are all labelled "average" (e.g. "Washing, 30 degrees C, average").
- The EF 3.1 dataset documentation (PRe Sustainability B.V.) explicitly states: ***"The different kind of detergent is not considered in this model."***
- Instead of modelling complete detergent formulations, the EF dataset uses a **single generic surfactant** (fatty alcohol sulfate from palm kernel oil, 0.0083 kg/kg laundry) as a simplified proxy for all detergent chemistry.

Advantages of our approach

Our modelling (based on Emelie Westall Lundqvist's work) provides **greater granularity and accuracy** compared to the PEFCR/EF approach:

Aspect	Carbonfact approach	PEFCR / EF 3.1 approach
Detergent types	Separate liquid and powder formulations with full ingredient lists	Single generic surfactant (fatty alcohol sulfate)
Ingredients modelled	9 ingredients (liquid) + 14 ingredients (powder) including packaging	1 surfactant only
Manufacturing energy	Differentiated: 0.25 kWh/kg (liquid) vs 0.69 kWh/kg (powder)	Not modelled separately
Impact differentiation	Shows +11-30% GHG difference between powder and liquid	No differentiation possible
Actionability	Enables product-level recommendations on detergent type	No guidance possible
Data sources	Kruschwitz et al. (2014), Toft Orbeck (2024), Sandin et al. (2019), Moazzem et al. (2021)	Kruschwitz et al. (2014), Quantis Data (2021)

This additional granularity is valuable for brands seeking to provide accurate care instructions and understand the environmental trade-offs of different laundering practices.

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7) Technical Machine Data

Machine Washing

AEG Washing machine, User Manual, LV7KR865E

Program	Load	Electricity consumption	Water consumption	Duration	Remaining moisture
Wool/Silk 30°C	1.5 kg	0.3 kWh	60 L	75 min	30%
Synthetics 40°C	3 kg	0.8 kWh	60 L	140 min	35%
Cottons 60°C	8 kg	1.3 kWh	80 L	220 min	44%

Passive electricity use of 0.3 W

Assumed load per program

Program	Average load factor (Kruschwitz et al. 2014)	Max weight (AEG Manual)	Assumed load per program
Wool	1.07	1.5 kg	1.07 × 1.5 = 1.605 kg
Synthetics	0.99	3 kg	0.99 × 3 = 2.97 kg
Cotton	0.68	8 kg	0.68 × 8 = 5.44 kg

Tumble Drying

AEG Dryer, User Manual, LD8KR845E

Program	Load	Spun at	Drying time	Electricity consumption
Cotton, closet dry	8 kg	1000 rounds/min	154 min	1.99 kWh
Cotton, iron dry	8 kg	1000 rounds/min	118 min	1.51 kWh
Synthetics, closet dry	3.5 kg	800 rounds/min	67 min	0.75 kWh

Passive electricity use of 0.13 W. All drying programs are eco.

8) Figures

Figure 1 — Climate Change (GWP100) bar chart

Horizontal bar chart ranking all 13 use-phase activities by GWP100 (kg CO2 eq per functional unit). Colour-coded by section.

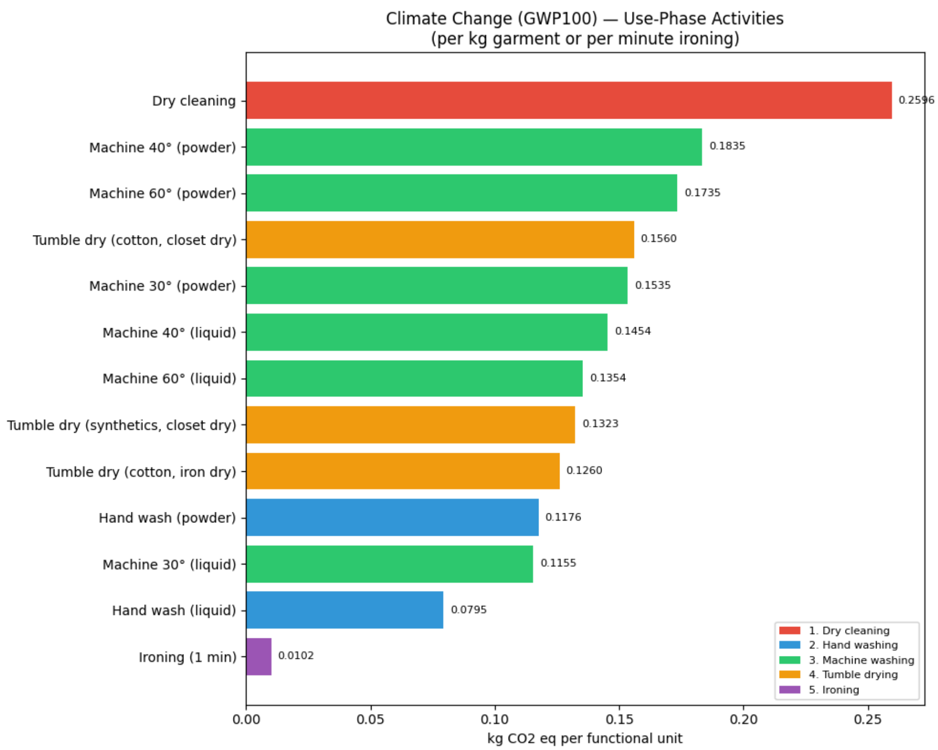
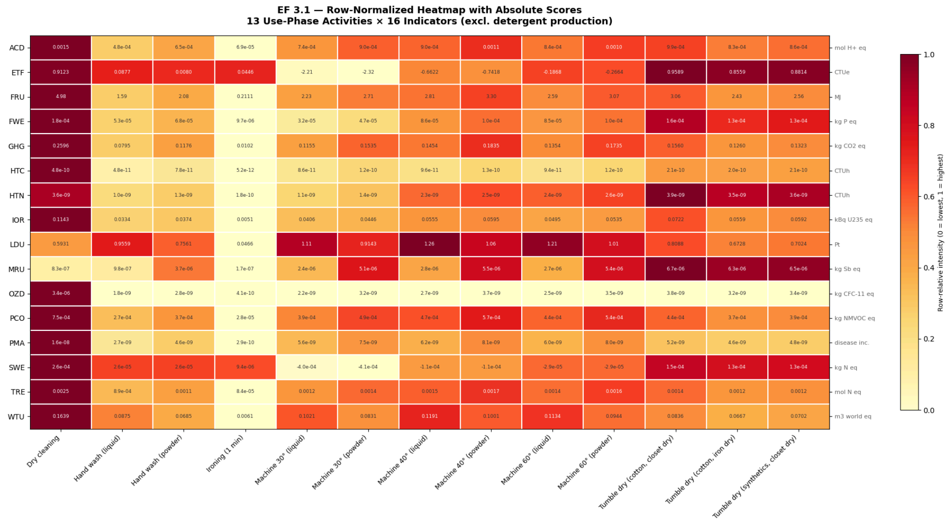


Figure 2 — Row-normalized heatmap (16 indicators × 13 activities)

Heatmap showing row-normalized intensities across all 16 EF 3.1 indicators, with absolute score annotations. Excludes detergent production activities.



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